

Quantifying LLM Practice Hours

A Multi-Method Calculation with Sensitivity Analysis

Noah Thomas Georgiana Chelednik

June 2026 — Revision 3 (refreshed from complete ChatGPT + Claude exports through 2026-04-21)

Executive Summary

This document provides a rigorous, auditable methodology for quantifying time invested in large language model interaction. Drawing from complete conversation exports (ChatGPT and Claude, both through April 2026, covering August 2023 – April 2026), I establish two defensible metrics:

Tier 1 — Documented In-Chat Interaction: ~3,700 hours

Baseline: 3,737 hours | **Range:** 3,177–4,298 | **Stress test floor:** 2,780

Time directly attributable to reading, writing, and processing within chat logs. Fully auditable from export data.

Tier 2 — Total LLM-Assisted Practice: ~6,500 hours baseline

Baseline: 6,465 hours | **Conservative range:** 5,101–7,435

In-chat interaction plus estimated offline work (implementation, iteration, research). Offline component is less auditable than Tier 1.

The methodology addresses potential objections through weighted reading models, explicit separation of in-chat versus offline time, and sensitivity analysis. Tier 1 survives aggressive stress testing. Tier 2’s “5,000+” framing is now supported even by the conservative floor: with the full ChatGPT corpus included, the conservative estimate (Tier 1 baseline + halved offline estimates) is 5,101 hours — above the “5,000+” claim rather than below it.

Revision note. This is the June 2026 revision of a methodology first published in December 2025 and revised in April 2026. The underlying logic, constants, and sensitivity parameters are unchanged across all revisions; only the underlying export data has been refreshed. This revision incorporates the complete ChatGPT export through April 2026 (delivered as 19 JSON shards), which resolves the gap noted in the April revision: the immediately prior ChatGPT export had omitted `conversations.json`, so the April revision held ChatGPT figures at their December 2025 values while refreshing only Claude. Both platforms are now current through April 2026. Relative to the April revision, the combined corpus grows from 65,400 to 71,427 messages and from 2,180 to 2,398 conversations, and all hour figures rise correspondingly.

1. Data Foundation

All calculations derive from complete conversation exports obtained directly from OpenAI (ChatGPT) and Anthropic (Claude) platforms via their official data export features. No sampling, estimation, or extrapolation was used — these are complete records.

Table 1: Combined ChatGPT and Claude usage metrics from complete exports.

Metric	Value
Total Conversations	2,398
Total Messages Exchanged	71,427
User Messages (Prompts Written)	35,225
Assistant Messages (Responses)	36,202
Unique Active Days	883
Date Range	2023-08-08 to 2026-04-21 (33 months)
User Words Written	2,061,417
Assistant Words Generated	12,982,012
Total Tokens Processed	23,392,755

1.1 Conversation Depth Distribution

Conversation depth indicates engagement quality. Brief exchanges suggest simple queries; extended conversations indicate sustained iterative work.

Table 2: Conversation depth distribution. Percentages are of the 2,363 conversations that contain extractable text messages; 35 conversations (Claude sessions with no text blocks) are excluded from depth and offline analysis.

Depth Category	Conversations	Percentage
Under 10 messages	1,173	49.64%
10 to 29 messages	622	26.32%
30 to 99 messages	411	17.39%
100+ messages	157	6.64%

Just over half (50.4%) of all conversations exceed 10 messages, with 568 conversations exceeding 30 messages. This depth distribution indicates sustained iterative engagement. Compared to the April 2026 revision, the 100+ message tier grew from 140 to 157 conversations and conversations exceeding 30 messages rose from 521 to 568 — consistent with continued long-horizon work and the inclusion of the full ChatGPT history.

2. Tier 1: Documented In-Chat Interaction Hours

Tier 1 captures time directly attributable to activity within the chat logs: reading AI responses, writing prompts, and cognitive processing between exchanges. This is the most auditable component — every input derives from verifiable export data.

2.1 Method 1: Weighted Reading Time

Users do not read all AI output at uniform depth. A weighted model accounts for variation: some content is skimmed, some read casually, some processed analytically.

Reading Speed Benchmarks

- Skimming / scanning: 400 words per minute (wpm)
- Casual reading: 200 wpm
- Analytical reading: 125 wpm
- Deep processing (technical content, code): 80 wpm

Weighted Distribution Model

The following distribution is applied to the 12,982,012 words of AI output. These proportions are estimates based on typical iterative LLM workflows; the sensitivity analysis tests robustness to different assumptions.

Table 3: Weighted reading time calculation.

Reading Mode	% of Content	Words	Hours
Skimmed / discarded	20%	2,596,402	108
Casual reading	35%	4,543,704	379
Analytical reading	30%	3,894,604	519
Deep processing	15%	1,947,302	406
TOTAL	100%	12,982,012	1,412

Method 1 Contribution: 1,412 hours

2.2 Method 2: Composition Time

Prompt writing differs from raw typing. Each prompt requires problem formulation, structural decisions, and often revision before sending.

- Raw transcription typing: 50 wpm
- Casual composition: 25 wpm
- Thoughtful / technical writing: 15–20 wpm

Total user words written: 2,061,417

Using a blended rate of 22 wpm (mix of quick queries and thoughtful prompts):

$$2,061,417 \div 22 \div 60 = \mathbf{1,562} \text{ hours}$$

Method 2 Contribution: 1,562 hours

2.3 Method 3: Inter-Exchange Cognitive Processing

Between receiving a response and writing the next prompt, practitioners evaluate output quality, identify iteration opportunities, and plan next steps.

Total user-assistant exchanges: 35,225 (one per user message)

Using a conservative average of 1 minute per exchange:

$$35,225 \times 1 \div 60 = \mathbf{587} \text{ hours}$$

Method 3 Contribution: 587 hours

2.4 Method 4: Session Overhead

Each active day involves overhead: opening the application, reviewing context, planning objectives, transitioning between threads.

Unique active days: 883

Using a conservative 12 minutes average daily overhead:

$$883 \times 12 \div 60 = \mathbf{177} \text{ hours}$$

Method 4 Contribution: 177 hours

2.5 Tier 1 Baseline Total

Table 4: Tier 1 component summary. The total is computed from unrounded component values; rows may not sum exactly due to rounding.

Component	Hours
Method 1: Weighted Reading Time	1,412
Method 2: Composition Time	1,562
Method 3: Inter-Exchange Processing	587
Method 4: Session Overhead	177
TIER 1 BASELINE	3,737

2.6 Tier 1 Sensitivity Analysis

To demonstrate robustness, I stress-test Tier 1 under deliberately pessimistic assumptions:

Table 5: Tier 1 stress test parameters.

Parameter	Baseline	Stress Test
% content skimmed	20%	40%
% deep processing	15%	5%
Composition speed	22 wpm	30 wpm
Processing time / exchange	60 sec	45 sec
Daily overhead	12 min	8 min

Stress Test Results

Table 6: Tier 1 under stress test assumptions.

Component	Stress Test Hours
Weighted Reading (40% skim, 5% deep)	1,076
Composition (30 wpm)	1,145
Processing (45 sec / exchange)	440
Session Overhead (8 min / day)	118
TIER 1 STRESS TEST FLOOR	2,780

Tier 1 Range Summary

Table 7: Tier 1 range summary.

Scenario	Hours
Stress test floor (pessimistic)	2,780
Conservative estimate (−15%)	3,177
Baseline calculation	3,737
Upper estimate (+15%)	4,298

Tier 1 Claim: ~3,700 hours (range: 3,177–4,298; stress floor: 2,780) — This represents purely in-chat, log-attributable time. The claim survives aggressive stress testing: even under simultaneously pessimistic assumptions on every parameter, the floor remains above 2,750 hours.

3. Tier 2: Total LLM-Assisted Practice Hours

Tier 2 extends beyond in-chat interaction to include offline work directly tied to LLM sessions. This component is less auditable than Tier 1 — the offline multipliers are estimates, not measurements. The sensitivity analysis tests robustness to different assumptions.

3.1 Offline Work Categories

Extended LLM sessions generate work outside the chat interface:

1. **Implementation:** Applying AI-generated code, structures, or recommendations to projects
2. **Testing / validation:** Running code, checking outputs, verifying recommendations
3. **External research:** Reading documentation or resources referenced in AI responses
4. **Iteration preparation:** Drafting or structuring complex follow-up prompts offline
5. **Synthesis:** Processing insights, connecting ideas across sessions

3.2 Offline Time Estimation (Baseline)

Offline time scales with conversation depth. The multipliers below are estimates based on typical workflow patterns for extended LLM sessions. They represent the least auditable component of this methodology.

Table 8: Baseline offline work hours by conversation depth.

Depth Category	Conversations	Offline Hrs / Conv	Total Hours
Under 10 messages	1,173	0.25	293
10 to 29 messages	622	1.0	622
30 to 99 messages	411	2.5	1,028
100+ messages	157	5.0	785
TOTAL	2,363	—	2,728

Baseline Offline Contribution: 2,728 hours

3.3 Tier 2 Sensitivity Analysis

Because offline multipliers are estimates, I stress-test with halved values:

Table 9: Stress test offline hours (50% of baseline multipliers).

Depth Category	Conversations	Stress Test Hrs	Total Hours
Under 10 messages	1,173	0.125	147
10 to 29 messages	622	0.5	311
30 to 99 messages	411	1.25	514
100+ messages	157	2.5	393
STRESS TEST TOTAL	2,363	—	1,364

Stress Test Offline Contribution: 1,364 hours

3.4 Tier 2 Total Calculation

Table 10: Tier 2 totals.

Component	Baseline	Conservative*
Tier 1: In-Chat Interaction	3,737	3,737
Offline Work	2,728	1,364
TIER 2 TOTAL	6,465	5,101

* *Conservative = Tier 1 baseline + halved offline estimates*

Tier 2 Range Summary

Table 11: Tier 2 range summary.

Scenario	Hours
Conservative (baseline Tier 1 + halved offline)	5,101
Baseline calculation	6,465
Upper estimate (+15% on baseline)	7,435

Tier 2 Claim: ~6,500 hours baseline (conservative floor: 5,101) — The “5,000+” framing reflects the baseline estimate and is now also satisfied by the conservative floor. Under conservative assumptions (halved offline multipliers), the total is 5,101 hours.

4. Validation Cross-Checks

4.1 Hours Per Active Day

Dividing totals by active days tests plausibility:

- Tier 1 baseline (3,737 hours) $\div$ 883 days = **4.23 hours / active day**
- Tier 2 baseline (6,465 hours) $\div$ 883 days = **7.32 hours / active day**

These figures are plausible. The most active day (1,130 messages on 2026-02-13) represents an outlier consistent with a prolonged multi-thread work burst; typical days range 2–7 hours.

4.2 Words Per Hour

Checking against combined word throughput:

- Total words (read + written): **15,043,429**
- At Tier 1 hours (3,737): **4,025 words / hour**
- Effective rate: **~67 wpm sustained** (reading + writing + processing)

This rate reflects analytical engagement — faster than deep study, slower than casual reading.

4.3 Literature Equivalence

For intuitive context (using 80,000 words as standard novel length):

- Words written: 2,061,417 = **25.8 novels** of original composition
- Words processed: 12,982,012 = **162 novels** of AI output
- Total engagement: **188 novels equivalent in 33 months**

5. Conclusion

This methodology establishes two claims with clearly defined scope and limitations:

Tier 1: "∼3,700 hours of documented in-chat LLM interaction"

- Baseline: 3,737 hours | Range: 3,177–4,298 | Stress floor: 2,780
- Fully auditable from export data
- Survives aggressive stress testing

Tier 2: "∼6,500 hours baseline of LLM-assisted practice"

- Baseline: 6,465 hours | Conservative floor: 5,101
- Includes in-chat interaction + offline work (implementation, testing, iteration)
- Offline component is estimated, not measured — less auditable than Tier 1
- "5,000+" reflects baseline; the conservative floor (∼5,100) also clears it

5.1 Claim Validity

Both claims are:

- **Empirically grounded:** Derived from complete, verifiable conversation exports
- **Methodologically transparent:** Every component calculation is explicit
- **Sensitivity-tested:** Both tiers include stress testing with pessimistic assumptions
- **Appropriately scoped:** Tier 1 and Tier 2 are clearly distinguished; limitations are acknowledged
- **Conservatively framed:** Claims state baseline figures with ranges, not upper bounds presented as certainties
- **Reproducible across revisions:** Constants, formulas, and sensitivity parameters are unchanged from the December 2025 and April 2026 revisions; only the underlying export data has been refreshed

5.2 Practical Context

The Tier 2 figure of ~6,500 hours represents approximately:

- **Over 3 years of full-time equivalent work** (at 2,000 hours / year) focused on human-AI interaction
- Sustained practice at a depth that exceeds typical professional AI adoption
- An empirical foundation for expertise claims in prompt engineering, AI workflow optimization, and recursive human-AI collaboration

Appendix A: Platform-Specific Data

Table 12: Platform comparison. Both platforms are now current through April 2026. Claude’s 563 conversations include 35 sessions with no extractable text blocks; its message, word, and depth figures reflect the 528 message-bearing conversations.

Metric	ChatGPT	Claude
Coverage Period	Aug 2023 – Apr 2026	Mar 2024 – Apr 2026
Conversations	1,835	563
Total Messages	50,412	21,015
User Messages	24,893	10,332
Assistant Messages	25,519	10,683
Active Days	847	292
User Words	1,330,677	730,740
Assistant Words	9,414,865	3,567,147
Avg Msgs / Conversation	27.47	39.80
Median Msgs / Conversation	8	16
Longest Conversation	872 msgs	637 msgs
Longest User Message	28,592 words	34,427 words
Most Active Day	494 msgs (2025-06-03)	1,130 msgs (2026-02-13)
Most Active Month	4,203 msgs (2025-04)	6,117 msgs (2026-02)

Appendix B: Calculation Formulas

Weighted Reading Time (Baseline)

$$\begin{aligned}
 \text{Hours} &= \frac{\sum (\text{Words} \times \text{Category}\%) \div \text{Speed}}{60} \\
 &= \frac{12.98\text{M} \times 0.20 \div 400 + 12.98\text{M} \times 0.35 \div 200 + 12.98\text{M} \times 0.30 \div 125 + 12.98\text{M} \times 0.15 \div 80}{60} \\
 &= \mathbf{1,412 \text{ hours}}
 \end{aligned}$$

Weighted Reading Time (Stress Test: 40 / 35 / 20 / 5 split)

$$\begin{aligned}
 &= \frac{12.98\text{M} \times 0.40 \div 400 + 12.98\text{M} \times 0.35 \div 200 + 12.98\text{M} \times 0.20 \div 125 + 12.98\text{M} \times 0.05 \div 80}{60} \\
 &= \mathbf{1,076 \text{ hours}}
 \end{aligned}$$

Composition Time

$$\text{Hours} = \text{Words} \div \text{Speed} \div 60$$

- Baseline (22 wpm): $2,061,417 \div 22 \div 60 = \mathbf{1,562}$ hours
- Stress test (30 wpm): $2,061,417 \div 30 \div 60 = \mathbf{1,145}$ hours

Inter-Exchange Processing

$$\text{Hours} = \text{Exchanges} \times \text{Minutes} \div 60$$

- Baseline (60 sec): $35,225 \times 1 \div 60 = \mathbf{587}$ hours
- Stress test (45 sec): $35,225 \times 0.75 \div 60 = \mathbf{440}$ hours

Session Overhead

$$\text{Hours} = \text{Days} \times \text{Minutes} \div 60$$

- Baseline (12 min): $883 \times 12 \div 60 = \mathbf{177}$ hours
- Stress test (8 min): $883 \times 8 \div 60 = \mathbf{118}$ hours

Offline Work (Baseline)

$$= (1,173 \times 0.25) + (622 \times 1.0) + (411 \times 2.5) + (157 \times 5.0)$$

$$= 293 + 622 + 1,028 + 785 = \mathbf{2,728} \text{ hours}$$

Offline Work (Stress Test: 50% multipliers)

$$= (1,173 \times 0.125) + (622 \times 0.5) + (411 \times 1.25) + (157 \times 2.5)$$

$$= 147 + 311 + 514 + 393 = \mathbf{1,364} \text{ hours}$$